



Normal Distribution

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NORMAL DISTRIBUTION Or GAUSSIAN DISTRIBUTION



- *Normal Distribution*
- *Standard Normal Distribution*
- *Normal curve*
- *Mean and Variance*
- *Standard Normal distribution table*

NORMAL DISTRIBUTION Or GAUSSIAN DISTRIBUTION



- *Normal Distribution is a continuous distribution of fundamental importance ,any quantity whose variation depends on random causes is distributed according to normal law ,its importance lies in the fact that a large number of distributions approximate to normal distribution .*
- *Independently Gauss and Adrain developed the formula for normal distribution and showed that errors fit well by this distribution .*

NORMAL AND STANDARD NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION



- Normal distribution, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean is more frequent in occurrence than data far from the mean. In graph form, normal distribution will appear as a bell curve.
- Normal distributions do not necessarily have the same means and standard deviations. *A normal distribution with a mean of 0 and a standard deviation of 1 is called a standard normal distribution.*

NORMAL DISTRIBUTION

➤ The continuous probability distribution with Probability Density function as follows :

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Note constants:

$\pi=3.14159$

$e=2.71828$

This is a bell shaped curve with different centers and spreads depending on μ and σ

where $-\infty < x < \infty$,

$-\infty < \mu < \infty$ and $\sigma > 0$

NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION



- It can be seen that the limiting form of the binomial distribution for large values of n when neither ' p ' nor ' q ' is very small, is the normal distribution.
- The Normal Distribution function $f(z)$ can be obtained where $n \rightarrow \infty$ and $p = q = \frac{1}{2}$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

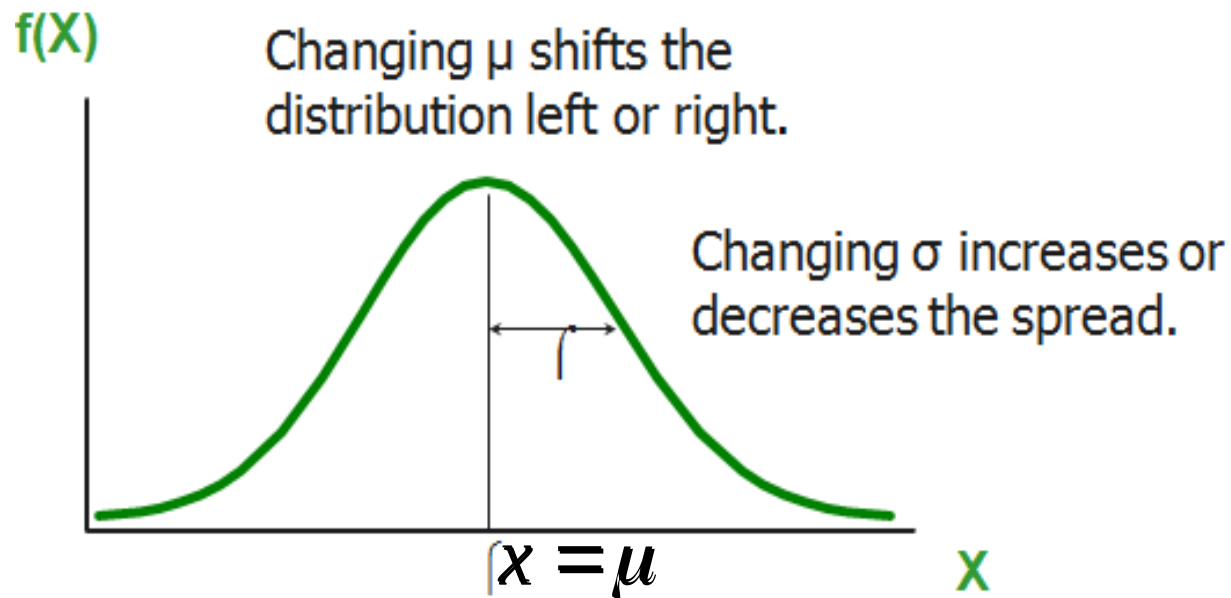
μ is mean

σ is standard deviation

NORMAL AND STANDARD NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION

➤ The graph of the probability function $f(x)$ is a Bell shaped curve symmetrical about the line $x = \mu$ and is known as normal probability curve



NORMAL AND STANDARD NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION



NORMAL DISTRIBUTION

➤ The line $x = \mu$ divides the total area under the Curve which is equal to 1 into two equal parts .

➤ The area to the right and left of the line $x = \mu$ is 0.5

It's a probability function, so no matter what the values of μ and σ , must integrate to 1!

$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1$$

STANDARD NORMAL DISTRIBUTION



STANDARD NORMAL DISTRIBUTION

$$E(X)=\mu = \int_{-\infty}^{+\infty} x \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$, \text{ Var}(X)=\sigma^2 = \int_{-\infty}^{+\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx - \mu^2$$

$$\text{Standard Deviation}(X)=\sigma$$

NORMAL AND STANDARD NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION

➤ $P(a \leq x \leq b) = \int_a^b f(x) dx$

➤ *In case of normal distribution*

$$P(a \leq x \leq b) = \frac{1}{\sigma\sqrt{2\pi}} \int_a^b e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

➤ *This integral cannot be evaluated by known methods of integration so we have to employ known techniques of numerical integration, which becomes tedious, so it must be standardized.*

➤ *taking,*

$$z = \frac{x - \mu}{\sigma} \quad \text{or} \quad x = \mu + \sigma z \quad ; \quad dx = \sigma dz$$

NORMAL AND STANDARD NORMAL DISTRIBUTION

FEATURES OF NORMAL DISTRIBUTION



➤ Let $z_1 = \frac{a - \mu}{\sigma}$ and $z_2 = \frac{b - \mu}{\sigma}$

be the values of 'z' corresponding to $x = a$ and $x = b$

$$P(a \leq x \leq b) = P(z_1 \leq z \leq z_2) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}} dz$$

➤ $F(z) = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{z^2}{2}} dz$ is same as

probability density function of normal distribution with mean $\mu = 0$ and $\sigma = 1$

➤ 'z' is called standard normal variate and $F(z)$ is called as standard normal curve which is symmetrical about the line $z = 0$

STANDARD NORMAL DISTRIBUTION



➤ $F(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ (Standard normal p.d.f.)

is the same as p.d.f. of normal distribution

with mean $\mu = 0$ and $\sigma = 1$ is the (Standard Normal p.d.f.)

$z = \frac{x - \mu}{\sigma}$ is called as the standard normal variate and

$F(z)$ is called as standard normal curve which is

symmetrical about the line $z = 0$

- *The curve is as follows :*
- *the integral represents area bounded by standard normal curve $F(z)$ between $z = z_1$ and $z = z_2$*

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \int_0^z e^{-z^2/2} dz \quad \text{represents area under the}$$

standard normal curve from $z = 0$ to z

- $\phi(z)$ denoted by $A(z)$ represents area (shaded portion)
- Since , total area is 1 , the area on either side of the $z = 0$ is 0.5
- Tabulated values which gives the area for different positive values of z are available and that helps us to in practical problems .

NORMAL AND STANDARD NORMAL DISTRIBUTION



Show that the area under the *N.D* curve is unity.

Proof:

Normal probability distribution is given by

Then the area A under the normal curve is

Put so

Or

NORMAL AND STANDARD NORMAL DISTRIBUTION

Continued.....



Put so

Or

Or

Here x , y are dummy variables

NORMAL AND STANDARD NORMAL DISTRIBUTION

Continued.....

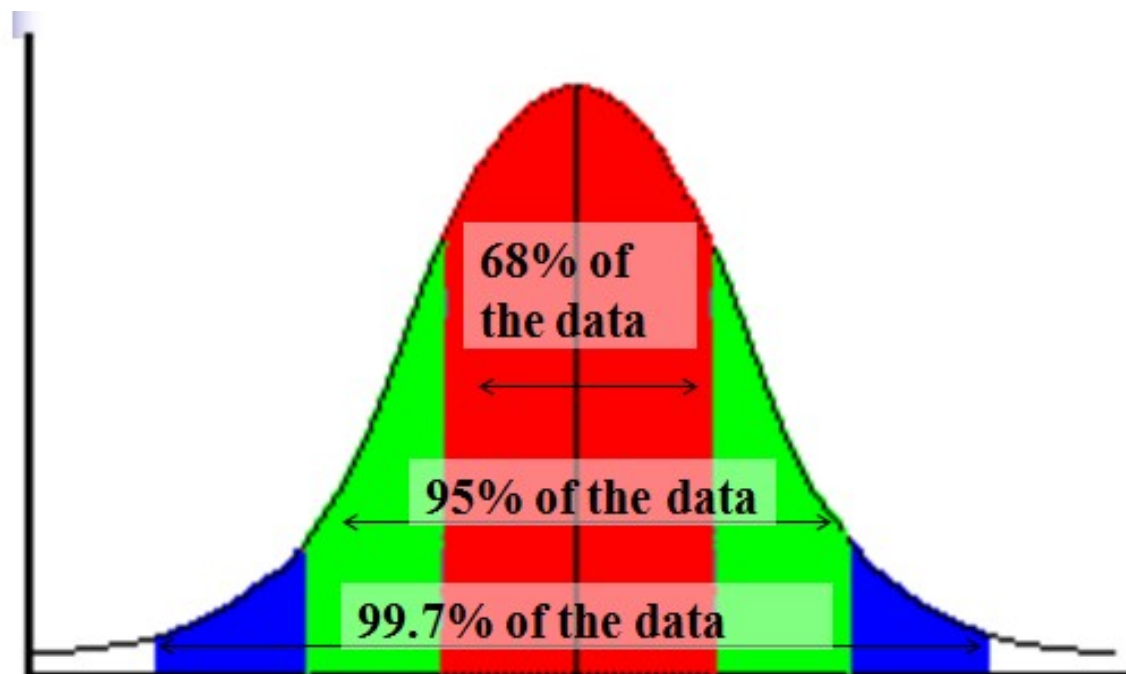


Put $J = \text{Jacobian} = \dots$

Limits for $\theta : 0 \text{ to } 2\pi$ (to cover the first quadrant $0 < x < \infty, 0 < y < \infty$)

So
 $= 1$

Thus $A = \text{area under the normal curve} = 1$.



Area under the normal curve is distributed as follows:

68.27% area lies between $\mu - \sigma$ to $\mu + \sigma$

i.e., between $\mu - \sigma$ and $\mu + \sigma$

94.45% area lies between $\mu - 2\sigma$ to $\mu + 2\sigma$

i.e., between $\mu - 2\sigma$ and $\mu + 2\sigma$

99.73% area lies between $\mu - 3\sigma$ to $\mu + 3\sigma$

i.e., between $\mu - 3\sigma$ and $\mu + 3\sigma$

Note: 50% area in the μ – interval ($-\infty$ to ∞)

99% area in the μ – interval ($-\infty$ to ∞)

No matter what μ and σ are, the area between $\mu - \sigma$ and $\mu + \sigma$ is about 68%; the area between $\mu - 2\sigma$ and $\mu + 2\sigma$ is about 95%; and the area between $\mu - 3\sigma$ and $\mu + 3\sigma$ is about 99.7%. Almost all values fall within 3 standard deviations.

$$\int_{\mu-\sigma}^{\mu+\sigma} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = .68$$

$$\int_{\mu-2\sigma}^{\mu+2\sigma} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = .95$$

$$\int_{\mu-3\sigma}^{\mu+3\sigma} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = .997$$

- $\phi(z)$ represents the area under the standard normal curve from 0 to z .
- Tabulated values which gives the area for different positive values of 'z' is readily available, such a table analogous to the format of a logarithmic table is called Normal probability table.
- $\int_{-\infty}^{\infty} \phi(z) dz = 1 \quad \Rightarrow \quad P(-\infty \leq z \leq \infty) = 1$
- $\int_{-\infty}^0 \phi(z) dz = \int_0^z \phi(z) dz = \frac{1}{2} \quad \Rightarrow \quad P(-\infty \leq z \leq 0) = \frac{1}{2}$
- $P(0 \leq z \leq \infty) \quad \text{or} \quad P(z \geq 0) = \frac{1}{2}$

$$P(-\infty \leq z \leq \infty) = 1$$

$$P(-\infty \leq z \leq 0) = \frac{1}{2}$$

$$P(0 \leq z \leq \infty) \quad \text{or} \quad P(z \geq 0) = \frac{1}{2}$$

$$P(-\infty < z < z_1) = P(-\infty < z \leq 0) + P(0 \leq z < z_1)$$

$$P(z < z_1) = 0.5 + \phi(z_1)$$

$$P(z > z_2) = P(z \geq 0) - P(0 \leq z < z_2)$$

$$P(z > z_2) = 0.5 - \phi(z_2)$$



THANK YOU

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